

**B.Tech. Degree I Semester Regular Examination November 2023****CE/ME/SE 23-200-0104A BASIC CIVIL ENGINEERING**

(2023 Scheme)

Time: 3 Hours

Maximum Marks: 50

**Course Outcomes**

On successful completion of the course, the students will be able to:

CO1: Summarize the types, uses and properties of various building materials.

CO2: Explain the different components of building and types of foundation.

CO3: Recognize the fundamental aspects and services in the field of civil engineering.

CO4: Discuss about the surveying techniques and to solve problems related with leveling.

CO5: Prepare site plan based upon the Kerala Panchayath and Municipality building rule.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

**PART A**(Answer *ALL* questions)

|    |                                                                                                                                         | (5 × 2 = 10) | Marks | BL | CO | PO     |
|----|-----------------------------------------------------------------------------------------------------------------------------------------|--------------|-------|----|----|--------|
| I. | (a) Illustrate the factors affecting workability of concrete.                                                                           |              | 2     | L2 | 1  | 1,2,12 |
|    | (b) Compare English and Flemish bond.                                                                                                   |              | 2     | L3 | 2  | 1,5    |
|    | (c) What are the principles of surveying?                                                                                               |              | 2     | L1 | 4  | 1,2    |
|    | (d) Explain the general provisions regarding the clearance from overhead electric lines as per the Kerala Municipal building rule 1999. |              | 2     | L3 | 5  | 1,8    |
|    | (e) List out any four features expected to be installed in intelligent building.                                                        |              | 2     | L2 | 5  | 1,8    |

**PART B**

(4 × 10 = 40)

|      |                                                                                                                                 |    |    |   |       |
|------|---------------------------------------------------------------------------------------------------------------------------------|----|----|---|-------|
| II.  | (a) Explain the classification of aggregates used for construction purpose.                                                     | 5  | L2 | 1 | 1,4   |
|      | (b) Construct a flow diagram showing the various stages in the manufacturing process of cement.                                 | 5  | L2 | 1 | 1,5   |
| OR   |                                                                                                                                 |    |    |   |       |
| III. | What are the various steps involved in making concrete?                                                                         | 10 | L1 | 3 | 1,4,9 |
| IV.  | What are the conditions under which a pile foundation is preferred? Also explain the classification of pile based on materials. | 10 | L1 | 2 | 1,2   |
| OR   |                                                                                                                                 |    |    |   |       |
| V.   | Compare stone masonry with brick masonry. Explain with neat sketches the classification of stone masonry.                       | 10 | L3 | 2 | 1,6,8 |

(P.T.O)

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|           |                                                                                                                                                                                                                                                                                                                                                                                                                                            | Marks | BL | CO | PO         |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----|----|------------|
| VI.       | Define surveying. Explain the field work done in surveying.                                                                                                                                                                                                                                                                                                                                                                                | 10    | L1 | 4  | 1,2,<br>10 |
| <b>OR</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |    |    |            |
| VII.      | The following are consecutive readings in metres taken using dumpy level and 3 m staff on a continuously sloping ground. The readings are 0.595, 1.240, 1.885, 2.385, 2.850, 0.230, 0.920, 1.930, 2.875, 0.465, 1.830 and 2.720. The first reading was taken on a bench mark of reduced level 192.115 m. Enter the above readings in a level book form and also calculate the reduced levels of points using height of collimation method. | 10    | L1 | 4  | 1,2        |
| VIII.     | Identify the possible ways in which green building can reduce or eliminate the adverse impact of buildings on the environment and occupant.                                                                                                                                                                                                                                                                                                | 10    | L3 | 5  | 1,5,<br>8  |
| <b>OR</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |    |    |            |
| IX.       | Outline the classification of roads based on (i) location and function (ii) materials and method of construction.                                                                                                                                                                                                                                                                                                                          | 10    | L2 | 3  | 1,6,<br>8  |

Bloom's Taxonomy Levels

L1 - 47%, L2 - 27%, L3 - 26%.

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